

The constant-speed question for the weighted spectral mean cited in arXiv:2009.10274 was answered by arXiv:2208.12407

Literature-status packet for run `fa_banach_001`

13 August 2026

Status: literature already answered affirmatively. Gan–Kim prove the exact constant-speed identity requested by Kim. The calculation below independently checks the match, but this packet is a retrieval result and makes no novelty claim.

Original signal and exact question

Let $\mathbb{P}$ be the cone of bounded strictly positive operators on a Hilbert space and let $\#$ denote the Kubo–Ando geometric mean. Sejong Kim, *Operator inequalities and gyrolines of the weighted geometric means*, arXiv:2009.10274 [1], defines

$$A \natural_t B = (A^{-1} \# B)^t A (A^{-1} \# B)^t, \quad d(A, B) = 2 \|\log(A^{-1} \# B)\|.$$

On source-PDF page 8, Remark 2(2), the paper calls the following a “challengeable problem”: prove, for all $s, t \in \mathbb{R}$,

$$d(A \natural_s B, A \natural_t B) = |s - t| d(A, B). \quad (1)$$

The published article phrases the parameters over $\mathbb{R}$; in particular this contains the originally extracted $s, t \in [0, 1]$ question.

Decisive later answer

Luyining Gan and Sejong Kim, *Revisit on spectral geometric mean*, arXiv:2208.12407 [2], explicitly announce the geodesic property in the abstract and introduction. Their Theorem 4.1 on source-PDF page 6 states exactly (1), for every $s, t \in \mathbb{R}$, and proves it using the composition law for weighted spectral means. This is an exact author-acknowledged answer: the supporting paper includes the author of the question, cites the 2021 paper, and proves the identical formula in a section devoted to the geodesic property.

Independent algebraic verification

There is also a short direct check. Put

$$C = A^{-1} \# B, \quad X_u = A \natural_u B = C^u A C^u.$$

The geometric mean $P \# Q$ is the unique positive solution Z of $ZP^{-1}Z = Q$. With $P = X_s^{-1}$ and $Q = X_t$, the positive operator $Z = C^{t-s}$ satisfies

$$ZP^{-1}Z = C^{t-s} X_s C^{t-s} = C^{t-s} C^s A C^s C^{t-s} = C^t A C^t = X_t.$$

Therefore

$$X_s^{-1} \# X_t = C^{t-s}.$$

Functional calculus now gives

$$d(X_s, X_t) = 2\|\log C^{t-s}\| = 2|t-s|\|\log C\| = |t-s|d(A, B),$$

which verifies both the later theorem and its exact fit to the source question.

Search evidence and scope

The run first searched its registry, ledger, solution, and attempt indexes for arXiv:2009.10274 and the phrases “spectral geometric mean”, “geodesic”, and “semimetric”; no prior local packet was found. A bounded external search on 13 August 2026 used the exact displayed formula, the source title, and the phrases “spectral geometric mean constant speed” and “geodesic semimetric”. It located arXiv:2208.12407 and its exact theorem. No novelty claim survives this match.

The answer applies to all bounded strictly positive operators and all real parameters, so no part of the stated question remains open. It does not make d a metric: the triangle inequality can still fail, as already noted in the source paper.

The packet stores the original article locally as `source_paper.pdf`. The answer article is stored as `supporting_paper_2208.12407.pdf`.

References

- [1] S. Kim, *Operator inequalities and gyrolines of the weighted geometric means*, Mathematical Inequalities & Applications 24 (2021), 491–514; [arXiv:2009.10274](#).
- [2] L. Gan and S. Kim, *Revisit on spectral geometric mean*, [arXiv:2208.12407v1](#) (2022), Section 4, Theorem 4.1.